

GODERICH, ON, Canada

WMO: 712610

Lat: 43.7647N

Lon: 81.7122W

Elev: 701

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	5.4	-5.0	4.3	1.8	-0.3	5.5	6.4	29.9	22.6	27.8	20.8	9.9	80	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	83.2	71.9	80.6	70.5	78.2	69.2	74.6	79.7	72.8	77.8	71.2	76.1	10.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	125.9	77.7	71.2	117.9	75.7	69.5	111.2	74.3	38.7	79.8	36.9	77.9	35.5	76.3	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.6	23.5	21.0	DB	-7.0	87.5	6.1	2.2	-11.4	89.1	-15.0	90.4	-18.4	91.7	-22.9	93.3	(4)
			WB	-7.4	77.5	6.1	2.5	-11.8	79.3	-15.4	80.8	-18.8	82.2	-23.2	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	23.8	24.3	31.4	42.7	54.2	63.4	67.4	66.5	61.1	50.8	39.7	30.3
	DBStd	17.81	10.37	10.16	10.56	8.89	8.71	7.38	5.70	5.51	7.02	8.02	8.08	8.16
	HDD50	3455	812	721	586	255	54	2	0	0	3	88	322	612
	HDD65	7133	1276	1141	1043	670	354	117	41	49	159	446	760	1077
	CDD50	2146	1	0	8	36	183	405	541	512	335	112	12	1
	CDD65	348	0	0	0	1	18	70	116	96	41	6	0	0
	CDH74	1840	0	0	1	6	115	425	613	449	211	20	0	0
	CDH80	279	0	0	0	0	14	78	92	69	25	1	0	0
Wind	WSAvg	10.4	12.9	11.8	11.0	11.1	9.5	8.2	7.5	7.8	8.8	11.1	12.6	13.0
Precipitation	PrecAvg	39.1	3.7	2.8	2.6	2.8	3.3	3.2	3.4	2.9	3.6	3.4	3.7	3.6
	PrecMax	50.2	6.1	4.9	5.9	5.4	6.9	7.0	6.1	6.3	9.4	6.6	6.6	7.1
	PrecMin	27.6	2.4	1.3	1.1	0.9	1.3	1.6	0.9	0.9	0.8	0.9	1.6	2.1
	PrecStd	5.9	1.0	1.0	1.2	1.2	1.3	1.4	1.3	1.1	1.8	1.4	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	53.3	66.6	73.6	81.8	86.1	86.1	85.9	83.3	76.8	64.1	55.6
		MCWB	51.4	50.4	58.5	62.6	67.7	73.0	74.9	73.9	71.3	66.2	56.5	53.4
	2%	DB	46.0	45.9	58.9	67.3	77.2	82.1	82.6	81.5	79.3	71.6	59.0	48.7
		MCWB	44.1	42.9	52.4	57.3	65.5	71.1	72.4	71.8	69.5	63.2	54.3	46.7
	5%	DB	40.5	40.6	51.5	62.4	73.2	78.9	79.8	78.6	76.2	67.4	55.4	44.6
		MCWB	38.7	37.7	46.6	53.4	62.7	68.7	70.7	70.3	67.6	60.9	51.6	42.7
	10%	DB	36.8	36.7	45.6	57.5	68.6	75.6	77.4	76.1	73.0	63.4	51.7	40.6
		MCWB	35.2	34.6	41.7	50.1	59.6	67.1	69.5	69.3	66.1	58.1	48.2	38.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	51.6	59.4	65.1	70.6	75.8	78.1	77.0	74.3	68.8	59.0	53.1
		MCDB	53.5	53.2	65.3	71.1	79.7	82.7	83.4	82.4	80.2	74.3	61.8	55.1
	2%	WB	44.5	43.2	53.3	59.0	67.1	72.8	74.7	74.1	71.4	64.9	55.2	47.2
		MCDB	45.5	45.1	58.0	64.7	74.5	79.6	79.5	78.6	76.2	69.7	58.0	48.6
	5%	WB	39.0	38.2	47.4	54.9	64.3	70.8	72.8	72.1	69.2	61.3	51.7	42.6
		MCDB	40.3	40.2	51.0	61.1	70.7	76.9	77.3	76.5	74.0	66.2	54.6	44.7
	10%	WB	35.3	34.8	42.1	50.7	61.0	68.5	71.1	70.5	67.1	58.4	48.4	38.5
		MCDB	36.5	36.4	45.5	56.8	67.1	73.8	75.6	74.6	71.9	62.9	51.5	40.4
Mean Daily Temperature Range	5% DB	MDBR	10.9	12.1	13.1	15.8	16.6	15.7	15.3	15.3	16.2	13.4	10.5	9.0
		MCDBR	13.8	15.2	20.0	23.3	20.8	18.9	18.0	17.4	19.0	18.4	15.8	13.1
		MCWBR	13.2	13.7	15.2	15.9	12.0	10.9	10.4	10.5	11.2	11.9	13.4	12.4
	5% WB	MCDBR	13.7	14.9	18.9	21.3	18.4	16.6	14.7	14.3	15.7	16.3	14.3	13.2
		MCWBR	13.5	14.0	15.5	16.4	12.7	10.6	9.9	10.1	10.8	11.7	13.1	12.8
Clear-Sky Solar Irradiance	taub	0.294	0.319	0.327	0.361	0.387	0.422	0.409	0.402	0.376	0.358	0.313	0.293	
	taud	2.410	2.274	2.360	2.333	2.326	2.253	2.323	2.324	2.392	2.411	2.508	2.461	
	Ebn at Noon	265	275	289	286	280	270	272	270	269	259	257	255	
	Edh at Noon	24	32	34	38	39	43	39	38	33	28	22	21	
All-Sky Solar Radiation	RadAvg	391	674	1100	1439	1728	1921	1905	1652	1307	781	448	288	
	RadStd	56	82	131	131	148	112	103	82	117	76	69	30	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	-1.51	-1.98	N/A	N/A	N/A	N/A
	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page